Interdicting Taiwan's Crude Oil Supply

Project summary | MA289-27-1 Independent Study | Cadet Dennis Pezan | Advisor: LTC Matthew Webb

THE PITCH

Taiwan imports nearly all of its crude oil by sea, and every barrel passes through a handful of straits. I am turning those shipping routes into a network model, and using LTC Webb's reinforcement-learning interdiction agent to ask two things: where would an adversary with limited resources act to cut Taiwan's oil supply, and can an AI trained on one kind of network reason about a very different one?

WHY IT MATTERS

Energy is widely identified as Taiwan's sharpest vulnerability in a blockade or quarantine. The chokepoints are not hypothetical: the Strait of Hormuz, which carried 61% of Taiwan's crude in 2025, has been largely closed since February 2026, and Taiwan has had to reroute supply. Most open analysis of a Taiwan blockade comes from wargames; this project brings an optimization model to the same question.

SCOPE

- **In:** crude oil only; the four regions that supply it (Persian Gulf, U.S. Gulf Coast, Oman, West Africa); the straits and canals it must pass; Taiwan's three refinery ports; one month of supply.
- **Out:** LNG and coal, detailed military campaigns, escorts and counter-attacks, and anything classified. It is a proof of concept built from open sources, not a targeting tool.

WHAT I AM LOOKING TO FIND

1. **Where the attacker acts.** Does a budget-limited attacker go after distant chokepoints such as Hormuz and Malacca, or Taiwan's own approaches?
2. **Quarantine versus blockade.** How does the answer change between a lower-cost coast-guard quarantine and a higher-cost naval blockade?
3. **Transfer.** Does an agent trained on the Ukrainian rail network still find sound strategies on a maritime network without retraining?

An early signal from the graph alone: the cheapest way to cut off all supply is a close cordon on Taiwan's west coast, because all three crude ports sit there.

PREVIOUS WORK IT BUILDS ON

- **Network interdiction:** the classic optimization models of Wood (1993) and Cormican, Morton and Wood (1998), surveyed in Smith and Song (2020).
- **LTC Webb's reinforcement-learning approach:** Webb and Caplice (2025), which trains an agent to find attacker strategies (defeat, isolate, canalize, divert) on the Ukrainian rail network.
- **Taiwan blockade analysis:** CSIS's quarantine scenarios and its 2025 "Lights Out?" blockade wargames, which identify energy shipping as the critical factor.
- **Data:** the U.S. Energy Information Administration's Taiwan brief (2026) for import sources and refinery capacity, and IMF PortWatch for shipping traffic.

APPROACH AND GOALS

The network is built: 55 points and 78 shipping lanes, each lane with the kind of action an attacker could take there (cyber, quarantine, blockade, economic pressure and others) and how costly and effective that action is. Next come baseline calculations, then runs of the agent on MIT SuperCloud with LTC Webb. The goals are:

- an open-source, defensible model of Taiwan's crude oil supply network;
- a clear picture of where interdiction is most effective at low, medium and high effort;
- an answer on whether the agent's learned strategy transfers from rail to sea;
- a poster presenting the results at the end of the semester.